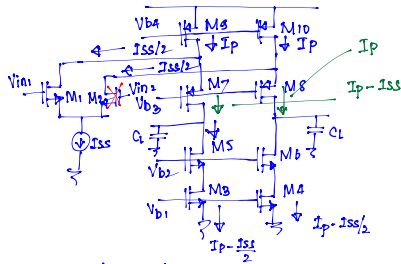


$$i = c \frac{dv}{dt}$$

$$\frac{dV_{out}}{dt} = \frac{I_{SS}}{2C_L}$$

$$\frac{dV_{out1}}{dt} = -\frac{I_{SS}}{2C_L}$$

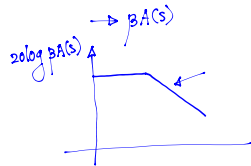
$$\left. \frac{dV_{out}}{dt} \right|_{\max} = \left. \frac{dV_{out1}}{dt} \right|_{\max} - \left. \frac{dV_{out2}}{dt} \right|_{\max} = -\frac{I_{SS}}{C_L}$$



Concept of stability

$$\begin{aligned} \frac{X_o(s)}{X_i(s)} &= \frac{A(s)}{1 + \beta A(s)} \\ &= \frac{A(s)}{1 + \beta(s)A(s)} \end{aligned}$$

loop gain $\rightarrow \beta(s)A(s)$
 $\downarrow$
 Independent of freq.



$$A(s) = \frac{A_M}{(1 + \frac{s}{\omega_{p1}})(1 + \frac{s}{\omega_{p2}})}$$

$$\begin{aligned} \beta A(s) &= \beta A(j\omega) \\ &= |\beta A(j\omega)| e^{<\beta(j\omega)} \end{aligned}$$

$$\text{At } \omega = \omega_1 \quad \left[\begin{aligned} |\beta A(j\omega)|_{\omega=\omega_1} &= 1 \\ <\beta A(j\omega) &= -180^\circ \end{aligned} \right]$$

Barkhausen's criteria

$$A_f(j\omega) = \frac{A(j\omega)}{1 + \beta A(j\omega)}$$

$$\text{at } \omega = \omega_1$$

$$A_f(j\omega) = \infty$$

$$A_f(j\omega) = \frac{A(j\omega)}{1 + \beta A(j\omega)}$$

$$a + jb = M e^{j\theta} = M (\cos\theta + j\sin\theta)$$

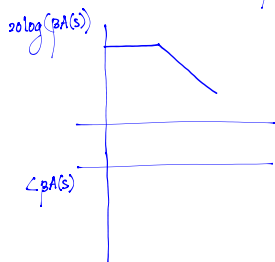
$$M = \sqrt{a^2 + b^2}$$

$$\theta = \tan^{-1} \frac{b}{a}$$

$$\tan^{-1} \left(\frac{b}{a} \right)$$

$$\beta A(j\omega) = A + j\beta = \sqrt{A^2 + \beta^2}$$

$$\theta = \tan^{-1} \left(\frac{\beta}{A} \right)$$



$$\frac{A}{(1 + \frac{s}{\omega_{p1}})(1 + \frac{s}{\omega_{p2}})}$$